SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	sticks and stones
Project / Solution Name	Resolve
Name (Leader)	Ee Jun Hao
Email & Phone (Leader)	ejhchesteree7@gmail.com 011-1615 1223
University Affiliation(s)	XIAMEN UNIVERSITY MALAYSIA
Live Prototype URL	https://resolve-sdc-hackathon-src.vercel.app
Code Repository URL (public / access granted to judges)	https://github.com/ejhchesteree7-code/Resolve-SDC-HACKATHON-

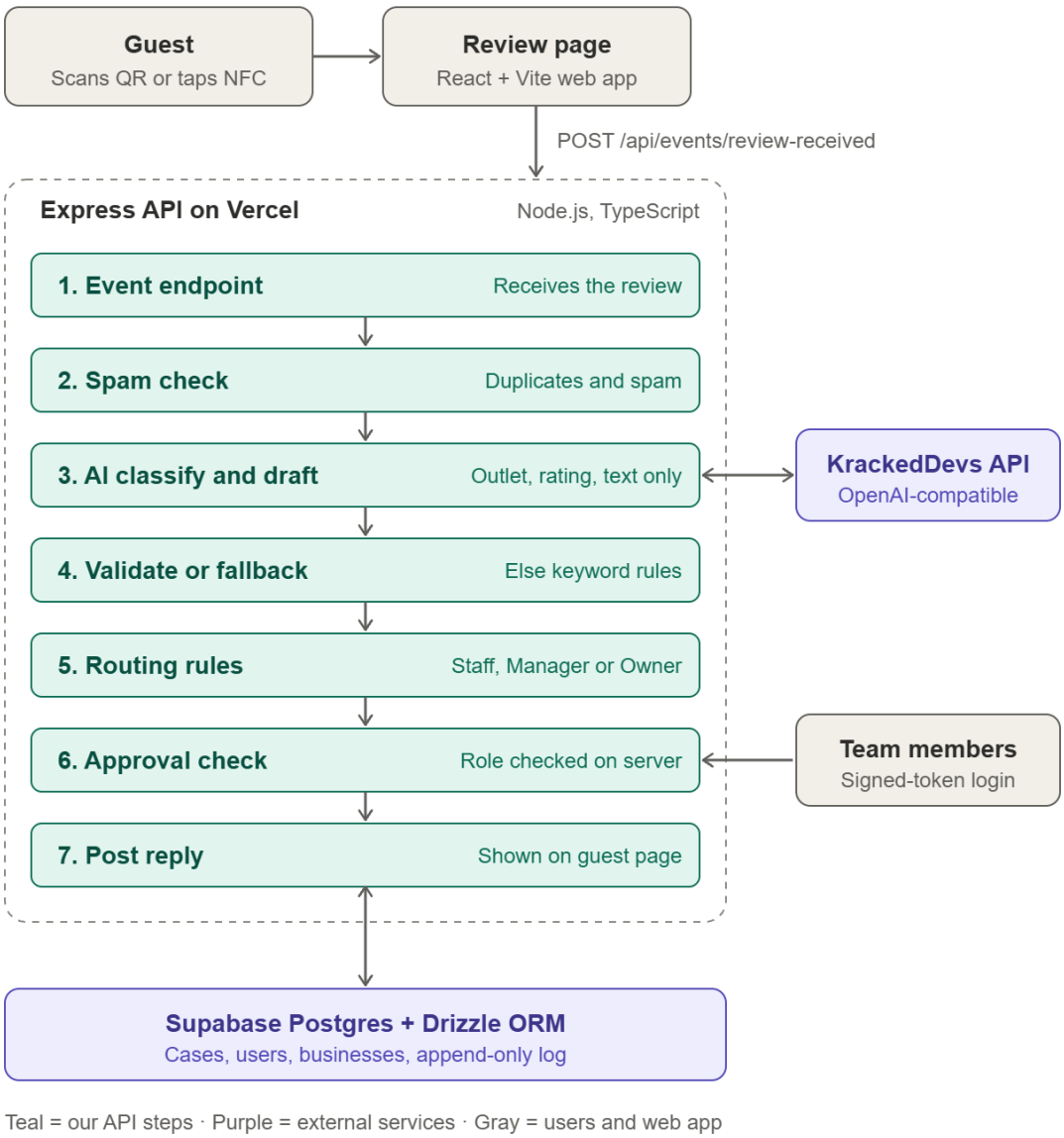
2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A cutomer scans an outlet's QR code or taps an NFC card, which opens the simulated review page in the React and Vite web app. The review is sent to the Express API (Node.js, TypeScript), deployed as a Vercel serverless function, through one event endpoint: POST /api/events/review-received. The API first runs a duplicate and spam check, then sends only the outlet, rating and review text to an OpenAI-compatible model through the KrackedDevs API. The model returns strict JSON (category, sentiment, severity, draft reply, reason), which the API validates if the call fails or times out after 12 seconds, built-in keyword rules take over. Plain routing rules assign the case to Staff, a Manager or the Owner. Cases, users, businesses and an append-only activity log are stored in Supabase Postgres through Drizzle ORM. Team members log in to the web app with signed session tokens, and every approval passes a server-side permission check before a reply is posted back to the guest page. See the attached architecture diagram.


Guest
Scans QR or taps NFC
Review page
React + Vite web app
POST /api/events/review-received
Express API on Vercel
Node.js, TypeScript
1. Event endpoint
Receives the review
2. Spam check
Duplicates and spam
3. AI classify and draft
Outlet, rating, text only
4. Validate or fallback
Else keyword rules
5. Routing rules
Staff, Manager or Owner
6. Approval check
Role checked on server
7. Post reply
Shown on guest page
KrackedDevs API
OpenAI-compatible
Team members
Signed-token login
Supabase Postgres + Drizzle ORM
Cases, users, businesses, append-only log
Teal = our API steps · Purple = external services · Gray = users and web app

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9	Main language (frontend and backend)	Apache-2.0
Node.js 22	Server runtime	MIT
React 19.1	Web app user interface	MIT
Vite 7.3	Frontend build tool	MIT
Tailwind CSS 4.3	Styling	MIT
wouter 3.13	Page routing in the web app	~~Unlicensed~~ Unlicense
TanStack Query 5	Loading and caching data in the web app	MIT
lucide-react 0.545	Icons	ISC
qrcode 1.5	Generating outlet QR codes	MIT
Express 5.2	Backend API	MIT
Drizzle ORM 0.45	Database queries and schema	Apache-2.0
node-postgres (pg) 8.23	Postgres connection	MIT
Zod 3.25	Validating API input and AI output	MIT
OpenAI Node SDK 7.39	Calling the OpenAI-compatible AI API	Apache-2.0

Name & Version	Purpose in Project	License
esbuild 0.28	Bundling the API for Vercel	MIT
pino 9.14	Server logging	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
KrackedDevs AI API (OpenAI-compatible, via litellm.rapidscreen.io)	Classify reviews and draft replies	Sponsor-provided
Google Business Profile API	Not called. Simulated by our guest page (planned for production)	None
WhatsApp Business Platform	Not called. Guest follow-up is simulated and logged	None

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosts the web app and the API (serverless function); auto-deploys from GitHub main	Hobby plan (free) / [North America]
Supabase	Postgres database (row-level security on)	Free [Asia Pacific (Singapore)]
GitHub	Code repository and pull requests	Free
Replit	Built the first version in Phase 1; backup deployment	[Core plan via SDCHACK26 promo / Free]

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-5.4-mini	KrackedDevs (OpenAI-compatible API)	Pre-trained, prompted only (no fine-tuning)	Provider terms of use
Basic mode (keyword rules and templates)	Built by the team	Rule-based fallback, not a trained model	Our own code

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Brew & Kin demo reviews (52 reviews)	Our repo: src/lib/db/seed.sql	Synthetic	Created by the team	N
Routing test set (55 labelled reviews)	Our repo: resolve-eval-set.ts	Synthetic	Created by the team	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Building the first version (Phase 1) that includes the frontend, API, database schema, first tests and in (Phase 2), it is used for generating pitch deck slides	Major
Claude (Anthropic), via Claude Code: Claude Opus 5.5	Finalist upgrades: logins, registration, Team page, new business types, duplicate check, overdue alerts, Impact page, generation of block and technical diagram,checking and drafting of documentation and report,	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every change went through a branch and a pull request that the team leader reviewed, tested on a preview deployment, and merged. Automated tests (38 in total) cover permissions, login, routing, duplicates, overdue rules and the Impact calculations, and must pass before merging. We test each feature by hand on the live site with all three roles, and a 55-review labelled test set checks routing accuracy on every Impact page refresh. The team can explain every part of the system and takes full responsibility for it.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
KrackedDevs	AI API credits (US$20 key per member)	Live review classification and reply drafting
Replit	[Core plan credits / Replit platform]	First version built with Replit Agent; backup deployment
RunPod	Not used	Not needed: one short AI call per review, no GPU work

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replaceable. Resolve calls the KrackedDevs key through the standard OpenAI-compatible interface, so any compatible provider can be swapped in by changing three environment variables, with no code change. The core workflow (routing rules, permissions, human approval and the locked activity log) never depends on the AI: if the API is unavailable, Basic mode uses keyword rules and reply templates, and every case still reaches the right person. The AI improves classification and draft quality; it is not required for the prototype to work.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

For customers, their first name or alias and a 1–5 rating and the review text. Team members: name, work email, role and a salted scrypt password hash (passwords are never stored in plain text). Only the outlet name, rating and review text are sent to the AI API; guest names and team data are never sent, and no data is used for model training. Data is stored in Supabase Postgres with row-level security on and no public access; the server is the only client. Access is limited by role and outlet, every action is logged, and removed staff lose access immediately. Secrets live in Vercel environment variables, never in the code. All data in the prototype is synthetic. In production, we would keep reviews for 12 months, then delete or anonymise them, in line with PDPA.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination or wrong category: the model must return strict JSON that is validated; invalid output falls back to keyword rules, and no reply is posted without human approval.
AI Bias or misreading Malay and Manglish: routing is decided by plain rules, not the model, and urgent topics (safety, harassment, money) always go to the Owner; a 55-review labelled test set tracks accuracy.
Provider outage, rate limits or exhausted credits: a 12-second timeout and Basic mode keep the workflow running.
Dependency on Single provider: the OpenAI-compatible interface lets us switch providers by configuration.
Spam or fake reviews: a duplicate and burst check flags them for a Manager.
Limitation: Google Business Profile and WhatsApp are simulated in the prototype.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 31/10/2026